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982 Extent of Medial Temporal Resection on Outcome from Anterior Temporal Lobectomy: A Randomized Prospective Study Operative Technique

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ABSTRACT: WE REPORT A prospective, randomized, blinded clinical trial comparing seizure and neuropsychological outcomes from anterior temporal lobectomies between two groups of patients. One group (n = 34) underwent hippocampal resection posteriorly to the anterior edge of the cerebral peduncle (partial hippocampectomy). In the other group (n = 36), the hippocampus was removed further to the level of the superior colliculus (total hippocampectomy). The amount of lateral cortical resection was the same between groups. Patients were followed-up for seizure outcome and neuropsychological morbidity. At 1 year postoperatively, the total hippocampectomy group had a statistically superior seizure outcome compared with the partial hippocampectomy group (69 versus 38% seizure-free), and examination of time to first seizure (survival analysis) revealed significantly superior outcomes associated with total hippocampectomy. There was no increased neuropsychological morbidity associated with the more extensive hippocampal resection.

KEY WORDS: Anterior temporal lobectomy; Clinical trial; Complex partial seizures; Epilepsy; Hippocampus; Memory

Anterior temporal lobectomy (ATL) is the most commonly performed surgery to control complex partial seizures, but it carries a variable success rate among different surgical centers^(11,26,37). Much of this variability is because no standardization exists for evaluating and deciding patients' candidacy for surgery. Even after identifying those patients who are suitable for ATL, little consensus exists concerning which temporal lobe structures need to be resected or how much resection should be done. For example, some surgeons tailor resections based on electrocorticography (ECoG) findings, whereas others routinely perform standard resections regardless of ECoG results^(32,35,46,59). Among surgeons who perform standard resections, no consensus exists concerning how much, or which, lateral or medial temporal lobe structures should be removed^(7,22,35,45,59).

There is a growing opinion, however, that optimal seizure outcome from ATL can be expected if the majority or all of the hippocampus is removed. This opinion has evolved from retrospective studies showing that patients who failed initial ATL improved after further hippocampal resection^(4,36,60,62) and also from retrospective analyses of nonrandomized patients^(3,5,31,44,47,49).

We now present data from a blinded, randomized, prospective, longitudinal clinical study in which patients with complex partial seizures from medial temporal lobes were randomly assigned to one of two surgical groups. At ATL, both groups had equal amounts of lateral temporal cortex excised. The only difference between groups was the degree of hippocampal resection. In one group, hippocampus was resected only to the level of the anterior margin of the cerebral peduncle. In the other, hippocampus was removed more posteriorly, to the level of the superior colliculus. Both groups were studied with respect to seizure and memory outcomes. The results were clear and consistent. Using various measures of seizure outcome, the group receiving the more aggressive hippocampal resection had the superior seizure outcome. Material-specific memory morbidity was not associated with the extent of hippocampal resection in either ATL group. The degree of verbal memory change after left ATL was associated with hippocampal pathological findings (memory deficits were seen in patients without well-formed hippocampal sclerosis).

PATIENTS AND METHODS

Patients

Patients who met the following criteria were considered eligible for randomization in this study: 1) medically intractable seizures; 2) nonretarded (Wechsler adult intelligence scale-revised full-scale intelligence quotient, >69); 3) chronological age between 18 and 40 years; 4) no magnetic resonance imaging (MRI) evidence of foreign tissue lesions; and 5) at least three seizures of unilateral mesial temporal lobe origin, as shown by long-term electroencephalogram/video recordings by subdural strip electrodes⁽⁶³⁾.

Patients meeting these criteria were informed of their eligibility for study participation at the end of long-term monitoring with strip electrodes. The nature, goals, and requirements of study participation were explained to the patients and their families. Their questions were answered, and they were given a copy of the informed consent (approved by both the University of Tennessee and the Baptist Memorial Hospital, Memphis, TN, IRBs) and allowed to make their decisions at home. They were not required to decide during initial discussions but were encouraged to call with any questions or concerns. When admitted for surgery (~3 wk later), the possibility of study participation was again reviewed with the patient and family, questions were answered, concerns were addressed, and the patients decided whether to participate. Of the eligible candidates, five declined to participate.

On the morning of surgery, the patients were

randomized to one of two types of surgery (described below). Patients and study team members other than the surgeon remained blinded to the type of surgery performed throughout the entire study.

Surgery

All of the patients underwent ATL under general anesthesia between January 1990 and August 1992. The surgery did not vary between dominant and nondominant hemispheres. After routine exposure of the temporal lobe, the anterior lateral 4.5 cm of temporal neocortex was removed, leaving the temporal horn of the lateral ventricle exposed along its anterior length. This dissection included the removal of superior through inferior temporal gyri, leaving hippocampus, parahippocampus, and fusiform gyri. Fusiform gyrus was then dissected piecemeal, leaving only hippocampal and parahippocampal gyri. A single silver clip was applied to the anterior tip of hippocampus, and then hippocampus was removed *en bloc* to either the anterior margin of the cerebral peduncle (partial hippocampectomy [P]) or to the level of the colliculi (total hippocampectomy [T]). The entire dissection of hippocampus was performed using the operating microscope. Therefore, the amount of lateral cortical resection was held constant for all patients, and the degree of hippocampal resection was systematically varied according to the randomization condition.

Two silver clips were applied to the posterior edge of hippocampus. By placing these single and double silver clips on the anterior and posterior edges, the specimen could be aligned after fixation to facilitate serial sectioning perpendicular to its long axis. After removal, the specimen was placed in formalin, fixed, and then serially sectioned perpendicular to its long axis, which, although it is somewhat curved, was mainly oriented along the anteroposterior axis, using previously described methods ⁽⁶¹⁾.

Hippocampal grading system

The hippocampal pathology grading system was based on reviewing serial sections of hippocampus removed *en bloc* (see description below). After reviewing an entire hippocampal specimen, those sections with the most severe pathological involvement were used for grading. Details regarding the development and refinement of this grading system are described elsewhere ⁽⁶¹⁾. The grading system yielded the following categories.

No hippocampal pathology present

None of the conditions which describe the following grades were met.

Grade 1: mild mesial temporal damage (MTD)

This grade applies to mild damage (gliosis with slight [$<10\%$] or no hippocampal neuronal dropout) involving sectors CA1, CA3, and/or CA4 of hippocampal pyramidal cell layer.

Grade 2: moderate MTD

This grade applies to moderate damage (gliosis with moderate [$10\text{--}50\%$] neuronal dropout) involving

sectors CA1, CA3, and/or CA4 of hippocampal pyramidal cell layer (if involvement is limited to CA3 and CA4, the lesion can be designated "end folium sclerosis").

Grade 3: moderate to marked MTD (hippocampal sclerosis or "classical" Ammon's horn sclerosis)

This grade applies to severe damage (gliosis with $>50\%$ neuronal dropout) involving sectors CA1, CA3, and CA4 of hippocampal pyramidal cell layer but sparing CA2, the resistant sector.

Grade 4: marked MTD (hippocampal sclerosis or "total" Ammon's horn sclerosis)

This grade applies to severe damage (gliosis with $>50\%$ neuronal dropout) involving all sectors of hippocampal pyramidal cell layer. The fascia dentata, subiculum, and parahippocampal gyrus may also be involved.

Also specified in the surgical report are the sectors of the hippocampal pyramidal cell layers (CA1, CA2, CA3, CA4), which are present and identifiable, the number of sections in which they are present, and whether they are present in continuity. The presence of layers in continuity provides a direct measure of how posterior the hippocampal resection went, because the layers anterior to the pes hippocampus are not in continuity.

Assessment of seizure outcome

The patients were typically seen by blinded study personnel for routine follow-up at 1, 3, 6, 9, and 12 months and yearly thereafter, and ratings of seizure outcome were determined at those times (seizure-free, not seizure-free). Thus, seizure outcome was judged by blinded personnel and not by the unblinded surgeon. At 6 months postoperatively, the patients were also interviewed by psychiatry and neuropsychology staff, and judgments regarding seizure outcome were independently garnered. Any disagreements regarding outcome were discussed, and in cases of disagreement, the patient was rated as not seizure-free. For assessment of surgical outcome at 1 year after surgery, the major focus of this article, two different project staff members (research nurse, neuropsychology fellow) blinded to type of surgery, independently contacted the patients and families of those rated as seizure-free to confirm their status. Every effort was made to verify the surgical outcomes of the patients.

Seizure outcome grading was rigorous and rigid. Patients were rated as seizure-free or not seizure-free. Seizure-free patients experienced absolutely no seizures (including auras) after surgery. Patients experiencing simple partial seizures (auras), complex partial seizures, or generalized seizures were rated as not seizure-free. Patients who had seizures after they decided on their own, and against medical advice, to stop taking or to reduce their anticonvulsant medications or had seizures in association with illnesses or other circumstances were rated as not seizure-free.

Neuropsychological assessment

The patients were given a complete neuropsychological assessment preoperatively and 6 months postoperatively. A wide variety of memory tests were administered to assess the effects of the extent of hippocampal resection on postoperative memory function. Administered memory tests included the following: Wechsler Memory Scale (WMS) ⁽⁵⁴⁾, Warrington Recognition Memory Test ⁽⁵³⁾, Rey Complex (as described in Lezak ^[28]), and the California Verbal Learning Test (CVLT) ^(8,9).

Russell's ⁽⁴⁰⁾ revision of the WMS was used, including immediate and 30-minute delayed recall of prose and figural stimuli. Immediate, delayed, and percent retention of the verbal and figural material represented the dependent measures from the WMS. The scoring system of Schwartz and Ivnik (Schwartz and Ivnik, unpublished manuscript) was used, as suggested by Russell ⁽⁴⁰⁾. From the Warrington Recognition Memory Test, the number of correct recognitions for the verbal (words) and visual (faces) stimuli were the dependent measures. The percent of visual material retained over a 15-minute delay was the dependent variable from the Rey Complex. The CVLT yields a large number of dependent variables. To minimize the number of statistical comparisons and still provide a theoretically meaningful examination of episodic verbal learning and memory, we used the results of a principal components analysis of preoperative CVLT data from 99 patients undergoing unilateral temporal lobectomies ⁽¹⁷⁾. A six-factor solution resulted, comparable to the factor structure described by Delis et al. ⁽⁹⁾. We selected two variables from each of the three factors. The first, and most powerful, factor, General Verbal Learning, accounted for 39% of the variance. Short-delay free recall had the highest loading. Total words recalled also had a high loading but was selected because of its clinical relevance. The remaining five factors, the proportion of variance they accounted for, and the primary variables selected for analysis were as follows: Acquisition Rate, 12% of the variance (consistency of free recall, learning slope); Response Discrimination, 9% of the variance (total intrusion and perseveration errors); Proactive Interference Effect, 8% of variance (List B recall, List B versus List A Trial 1); Serial Position Effect, 6.6% of variance (percent of recall from primacy and recency regions of the list); and Learning Strategy, 5.5% of variance (serial and semantic clustering).

The preoperative minus postoperative difference score was computed for each index and subjected to analysis of variance. The predictor variables were the extent of hippocampal resection (partial versus total), degree of hippocampal sclerosis (no/mild [Grades 0-2] versus moderate/severe [Grades 3-4]), and the interaction of extent of resection by degree of sclerosis. The degree of hippocampal sclerosis was included as a predictor variable, because we had previously found it to be a significant predictor of memory outcome after ATL ⁽²⁰⁾. Analyses were computed separately for the left and right ATL groups.

RESULTS

Efficacy of randomization

As shown in *Table 1*, there were no differences between the partial and total hippocampectomy groups in terms of the following variables: chronological age ($P = 0.40$), age at onset of epilepsy (*Table 1*) ($P = 0.10$), gender ($P = 0.64$), laterality of surgery ($P = 0.67$), speech lateralization (left, right, bilateral) ($P = 0.38$), presence/absence of hippocampal sclerosis determined by either a binary system (present/absent) ($P = 0.49$) or by individual grades of hippocampal sclerosis (Grades 0-4) ($P = 0.59$). The randomization process was clearly successful in equating groups in terms of potential confounding variables.

Effects of type of surgery

Outcome at 1-year follow-up

At 1-year follow-up, there was a significant effect of type of surgery ($\chi^2 [1] = 6.86, P = 0.009$). Sixty-nine and four-tenths percent of total hippocampectomy patients, compared with 38.2% of partial hippocampectomy patients, were completely seizure-free.

A logistic regression analysis was computed, with the dependent variable being surgical outcome (seizure-free, not seizure-free) (*Table 2*). Independent variables included extent of resection (*Table 2*) (partial versus total hippocampectomy), age at surgery, age at onset of epilepsy, laterality of surgery, gender, and presence/absence of hippocampal sclerosis. Only the extent of resection (total hippocampectomy) was a significant predictor of outcome ($\chi^2 = 6.69, P = 0.02$, odds ratio = 4.16). There was a trend ($P = 0.07$) for younger patients to have more favorable outcomes. This analysis included 61 patients, because pathology specimens were unobtainable for 9 patients. The above analysis was recalculated, deleting the hippocampal sclerosis variable so that all 70 patients could be included. Again, only the extent of resection was statistically significant ($\chi^2 = 6.99, P = 0.008$, odds ratio = 4.12), with better outcome associated with total hippocampectomy.

Survival analyses

The partial and total hippocampectomy groups had minimum follow-ups of 1 year, with maximum follow-ups of 3.94 years for the partial and 3.99 years for the total hippocampectomy groups. To take advantage of this extended follow-up, survival analyses were calculated for time to first seizure. When seizures occurred, they appeared earlier in the partial hippocampectomy group than the total hippocampectomy group (1.3 versus 1.9 yr). *Figure 1* shows the survival function estimates for the two groups. As can be seen, the partial hippocampectomy (*Fig. 1*) group continues to have patients experience postoperative seizures and does not level off until ~12 months after surgery. On the other hand, the total hippocampectomy group survival curve levels off from 4 to 6 months postoperatively. The difference between survival curves is statistically significant (log-rank test, $\chi^2 [1] = 4.05, P = 0.04$).

Memory outcome

Material-specific memory effects after left or right ATL were examined (Table 3). For patients undergoing right ATL, there were no significant findings for any memory measure. Neither the extent of hippocampal resection, degree of underlying hippocampal sclerosis, nor their interaction was associated with visual memory outcome.

For patients undergoing left ATL, several trends were evident. First, the extent of hippocampal resection (partial versus total) (Table 3) was not associated with significant pre- to postoperative change on any verbal memory index. Second, the degree of hippocampal sclerosis was predictive of verbal memory outcome after left ATL. In each of the instances that was statistically significant, poorer verbal memory outcome was associated with resection of hippocampus, revealing no/mild sclerosis. On the CVLT, resection of hippocampus with no/mild sclerosis was associated with a significant reduction in the total number of words recalled across trials, poorer short-delay free recall, and reduced slope of the verbal learning curve (Fig. 2). On the WMS, there was decreased immediate and delayed recall of prose. It is important to emphasize that there were no main effects of the extent of hippocampal resection nor any interaction between the extent of resection and the degree of hippocampal sclerosis.

Complications

There were complications in five patients (7%) as follows: two patients with partial hippocampectomy (one subgaleal cerebrospinal fluid fistula and one temporary diplopia) and three patients with total hippocampectomy (temporary IIIrd nerve paresis, two with cerebrospinal fluid subgaleal fistula).

DISCUSSION

Complex partial seizures are one of the most difficult forms of epilepsy to successfully treat with presently available antiepileptic drugs⁽⁴²⁾. The majority of patients with complex partial seizures have seizure foci residing within one temporal lobe. If they are refractory to medical treatment, they are often excellent candidates for ATL^(56,57). However, seizure and neuropsychological outcomes from ATL may vary widely among different surgical centers^(11,26,37).

Variation in surgical approach

A major factor contributing to this outcome variability is that no clinical trial data are available to objectively support which of several surgical techniques is best suited for the typical patient. Moreover, it is unclear which temporal lobe structures need to be removed for optimal seizure control in nonlesional (nonforeign tissue) cases, even when volumetric MRI shows typical mesial temporal sclerosis.

Techniques vary within a spectrum of approaches that include performing surgery under local or general anesthesia, guiding resection by acute or

chronic ECoG data^(1,27,33), or simply performing one of several standardized operations without regard to ECoG^(36,59). Even when considering standardized resections, there is no consensus concerning which structures, or how much of any one structure, should be removed for optimal seizure control. Some surgeons include superior temporal gyrus⁽⁵⁹⁾; others do not⁽⁴⁵⁾. Weiser and Yasargil⁽⁵⁵⁾ reported selective resection of only amygdala and hippocampus, whereas Keogan et al.⁽²⁵⁾ reported good results from resecting only lateral temporal cortex, leaving hippocampus intact. Recently, Goldring et al.⁽¹²⁾ hypothesized that the common denominator to account for a similarity in ATL outcomes from various surgical approaches was the resection of entorhinal cortex.

Hippocampal resection and seizure outcome

In the past decade, evidence has grown stronger that resection of hippocampus plays an important role in determining outcome from routine ATL. Several retrospective reports suggest that at least 50% of patients not having optimal seizure outcome from ATL may improve with further hippocampal resection (assuming the initial ATL left a significant portion remaining)^(4,36,60,62). Based on long-term depth electrode recording (inserted in a posterior to anterior trajectory along the long axis of the hippocampus), Spencer et al.⁽⁴⁷⁾ hypothesized that 20% of patients undergoing ATL have the primary epileptogenic potency within posterior, not anterior, hippocampus. Therefore, they suggested that seizure outcome would improve with a more complete hippocampal resection than is achieved with the standard *en bloc* technique.

Until now, the hypothesis presented by Spencer et al.⁽⁴⁷⁾ has not been tested with a randomized, blinded, prospective clinical trial. In this study, we prospectively randomized patients into two groups; the first group received a standard hippocampal resection measured anatomically as extending to the anterior edge of the cerebral peduncle (partial hippocampectomy), and the other group underwent hippocampal resection further posteriorly to the superior colliculus (total hippocampectomy). In both groups, the amount of lateral cortex was held constant at 4.5 cm. Therefore, the only difference between the two groups was the extent of medial resection.

Design and outcome of clinical trial

The study group of 70 patients was very homogeneous because of the strict entrance criteria. All patients had a full-scale intelligence quotient >69, no MRI evidence of foreign tissue lesions as a possible cause of their epilepsy, and complex partial seizures originating electrically from the mesial temporal region unilaterally, as proven by long-term ictal monitoring with implanted bilateral subdural strip electrodes, using previously described techniques⁽⁶³⁾. For the majority of patients, hippocampal sclerosis was evidenced by the pathological examination of removed tissue.

The study group patients were randomized to the type of surgical procedure, and examination of the

comparability of the partial and total hippocampectomy groups showed the randomization process successfully equated the groups on all important variables (laterality of ATL, gender, chronological age, age of onset of epilepsy, duration of seizures, speech dominance, presence of hippocampal sclerosis). All patients and study personnel (except the surgeon) were blinded to each patient's randomization condition, and seizure outcomes were verified several times by multiple project staff.

The data clearly showed that the group undergoing more extensive hippocampal resection (total hippocampectomy) had the superior seizure outcome. This was demonstrated in several ways, including the following: 1) examination of the proportion of patients rendered completely seizure-free (64 versus 38%) at 1 year postoperatively; 2) two logistic regression analyses, which included additional predictor variables but in which only the extent of resection (total hippocampectomy) was a significant predictor of seizure-free outcome; and 3) examination of the distribution of time to first seizure throughout the entire period of follow-up, which demonstrated a significant difference in the survival curves with earlier time to first postoperative seizure and a greater proportion of not seizure-free outcomes in the partial hippocampectomy group. The importance of complete seizure freedom after ATL is based on the observation that seizure-free patients show the sharpest psychosocial gains and are improved, even as compared with those who exhibit only simple partial seizures^(16,18,19,50-52).

Memory outcome

A potential concern regarding extensive hippocampal resection is the risk of postoperative memory dysfunction. Although material-specific memory deficits after ATL have been described since the 1950s (verbal memory decline after left ATL and, less reliably, visual memory decline after right ATL), a number of studies have suggested that the risk of postoperative memory decline is associated with more extensive hippocampal resection⁽²³⁾. However, these studies have typically involved only postoperative memory performance assessment, not pre- to postoperative *change* in performance, and the type of surgery and/or extent of hippocampal resection was not randomly determined. It remains possible that these studies have confounded pre- with postoperative memory impairments and/or that other factors are responsible for the memory outcome results. In fact, several recent investigations have examined the degree of pre- to postoperative memory change associated with varying degrees of mesial temporal resection^(24,29,58) or have compared memory outcomes of temporal lobectomy (with partial hippocampectomy) versus selective amygdalo-hippocampectomy⁽¹³⁻¹⁵⁾. These studies consistently have not found a relationship between memory outcome and extent of hippocampal resection. Again, none of these studies involved patients randomized to different surgical conditions. Our results support these studies in that in no instance was verbal

memory change associated with the extent of left hippocampal resection nor was the extent of right hippocampal resection associated with the degree of visual memory change.

That verbal memory change (decline) was associated with a lack of hippocampal sclerosis is consistent with several previous studies, indicating that verbal memory loss is associated with quantitative and qualitative histopathological^(20,21,34,38,41) and MRI volume⁽⁴⁸⁾ evidence of relatively normal hippocampal structure. These relationships were noted on the most powerful CVLT factor (General Verbal Memory) as well as the Logical Memory subtest of the Wechsler Memory Scale. That visual memory change after right ATL is a less reliable outcome, and more difficult to predict, has been noted by several previous investigators^(2,6,10,30,39).

CONCLUSION

We report the first prospective, blinded, randomized clinical trial to compare the extent of hippocampal resection on seizure and memory outcome from ATL. Better seizure control resulted from aggressive hippocampectomy, with no increased damage to memory functions. When memory declines did occur, they were in patients without pathological evidence of left hippocampal sclerosis and in whom seizure onset was at a later age.

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COMMENTS

The randomized clinical trials of surgical procedures has been a controversial and too infrequently visited subject. As surgeons, we historically have often individually modified procedures based on hypothesis gleaned from anecdotal evidence. Proof of efficacy and safety often wait until a substantial number of patients have been treated and then compared with standard treatment in a retrospective analysis. Retrospective studies are always subject to bias and too many uncontrolled variables; therefore, adequate proof that the procedure is the best is not forthcoming and real data to guide practitioners is obfuscated by a controversial literature. The procedure in question, however, often becomes so firmly entrenched among its biased supporters that it is never studied prospectively in randomized clinical trials.

Neurosurgery is replete with examples of entrenched surgical procedures which emerged from a biased hypothesis, and the conduct of temporal lobectomy is a good example. Based on our depth electrode recordings from medial temporal structures and historical evidence regarding the pathology of mesial temporal sclerosis corroborated by pathological findings, we began in the late 1970s and early 1980s to perform radical resection of medial temporal lobe structures, including the amygdala, hippocampus, and parahippocampus and sparing the lateral temporal neocortex, except for the pole.

As in the theoretical procedure described above, we became so confident that this was the best procedure for the temporal lobe epilepsy, we were unable to overcome our bias and perform proper randomized clinical trials.

Wyler et al. have done just that. This is a unique study that may not ever be repeated. Magnetic resonance imaging emerged as a powerful tool during the conduct of this study. It allows preoperative assessment of medial temporal atrophy that correlates well with the pathology of mesial temporal sclerosis. This has obviated the need for intracranial recordings in selected patients. However, all of the patients in this study had the additional standard of invasive electrophysiology applied to the selection process.

Although partial seizures (the aura) were not examined separately, the study clearly demonstrates that patients with medial temporal lobe epilepsy have better seizure control with more extensive medial (not lateral) resection and do not suffer additional cognitive deficits for this resection. This study corroborates other reports, including our own, that postoperative verbal learning problems sometimes seen after dominant temporal lobectomy are not caused by the extent of the medial resection but on the presence or absence of pathological findings, i.e., mesial temporal sclerosis.

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This manuscript addresses a very important

question regarding the length of hippocampus that should be resected in patients undergoing surgery for mesial temporal lobe epilepsy. The authors' findings, that there is a significant difference in seizure outcome between limited and more extensive hippocampal resection and that there does not seem to be a neuropsychological price to be paid for more extensive resection, are major contributions. Although one might have liked an additional measure or documentation of the length of hippocampus resected (such as absolute measurement intraoperatively, pathologically, or radiologically) and although the power of the methodology to have detected a meaningful neuropsychological change may not be explicit in the text, the study seems to have been very well done overall. The study design for this surgical treatment question, prospective, randomized, and blinded except for the nonevaluating surgeon, is particularly appropriate and noteworthy. The growing body of evidence in favor of more extensive hippocampal resection receives strong support in this clear and practical report.

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The authors present a well-controlled prospective clinical trial, comparing partial and total hippocampectomy. Such trials are difficult to perform and are rarely undertaken to resolve controversies in the field of epilepsy surgery. Unfortunately, however, although the data indicate significantly fewer seizures after total hippocampectomy, the inclusion of auras as seizures in this study does not permit these data to be used to resolve the critical clinical issue regarding postoperative improvement. Almost all epilepsy surgery programs today discount nondisabling auras as seizures, and only those postoperative ictal events that impair function are of practical concern. Despite a recent report that the quality of life for patients with postoperative auras may not be as good as that for patients who are completely seizure-free, there remains a major difference between those with auras only and those with residual complex partial or generalized seizures (i.e., the former have no limitations on such activities as driving an automobile). It would be important to know, therefore, whether the apparent superiority of total hippocampectomy noted in this study could be accounted for entirely by the persistence of auras in the partial hippocampectomy group. Furthermore, by failing to distinguish between auras and complex partial or generalized seizures, the authors have missed a potential opportunity to explain the pathophysiological basis of a puzzling observation; postoperative auras are more likely to persist in patients with hippocampal sclerosis than in those with other epileptogenic lesions. If auras had been identified separately, it might have been possible to demonstrate that residual auras in these patients reflected a failure to resect damaged tissue in the posterior hippocampus.

Jerome Engel, Jr.

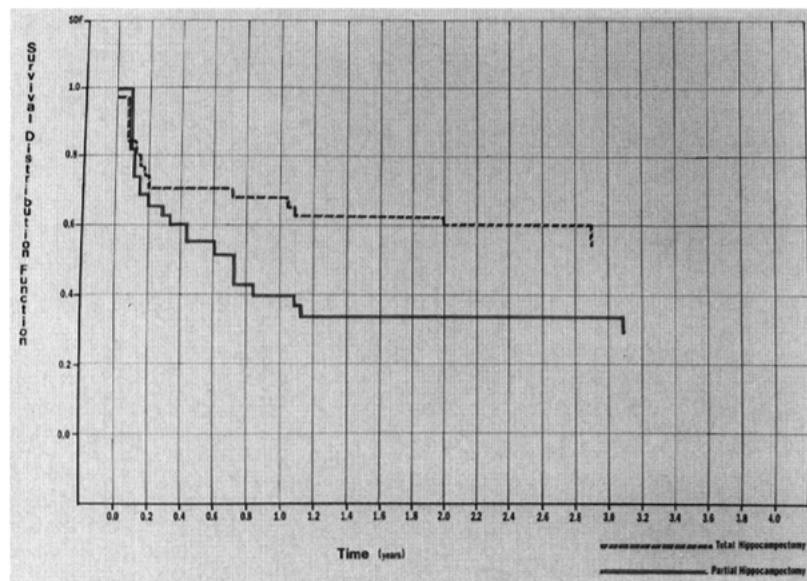


Figure 1. Survival curves depicting time to first seizure in the partial and total hippocampectomy groups. Seizure occurrence is represented by downgoing notches in the respective curves, and the horizontal axis represents time since surgery.

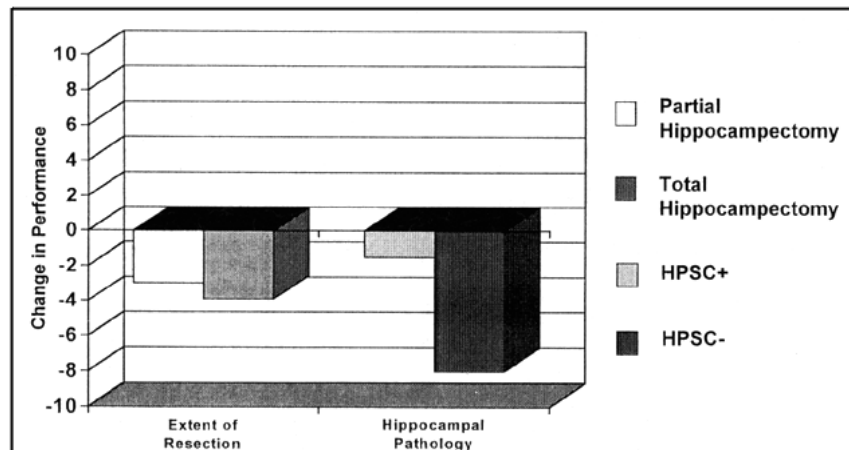


Figure 2. Pre- to postoperative change in the number of words recalled across five learning trials in the left ATL group. Patients were first examined according to extent of hippocampal resection (partial versus total), and there was no significant difference between groups. Patients undergoing left ATL were then reclassified according to presence or absence of hippocampal sclerosis. Adverse memory outcome was associated with resection of nonsclerotic left hippocampus (*HPSC*-).

	Degree of Hippocampal Resection	
	Partial (n = 34)	Total (n = 36)
Age at surgery	30.5	31.2
Age at onset of epilepsy	11.3	9.8
Age at follow-up	33.2	33.8
Gender	17 F/17 M	20 F/16 M
Laterality of resection	22 L/12 R	25 L/11 R
Cerebral language representation	33 L/1 R	32 L/3 R/1 B
Hippocampal sclerosis ^b	8 no, 18 yes	8 no, 27 yes

^a L, left; R, right. Numbers are means.

^b Hippocampal specimens could not be obtained or could not be reliably graded for some patients. The number of patients represented here is therefore lower than the total number.

Table 1. Patient Characteristics^a

Predictor Variables	χ^2	P Value	Odds Ratio
Type of surgery	5.7	0.02	4.2
Age at surgery	3.3	0.07	0.9
Age at onset	0.03	0.87	1.0
Laterality of surgery	0.16	0.69	1.3
Gender	1.0	0.32	1.8
Presence/absence of hippocampal sclerosis	0.72	0.39	2.0

Table 2. Results of Logistic Regression Analysis

	Extent of Hippocampal Resection		Hippocampal Sclerosis		Effect of Extent of Hippocampal Resection	Effect of Degree of Hippocampal Sclerosis
	Partial	Total	Absent	Present		
Left Anterior Temporal Lobectomy						
California Verbal Learning Test						
Total words	-3.0 (8.9)	-3.95 (7.1)	-8.0 (6.7)	-1.5 (7.98)	ns	0.03
Short delay	-2.0 (2.9)	-1.1 (3.9)	-3.8 (3.9)	-0.4 (3.1)	ns	0.01
Semantic clustering	-1.1 (0.6)	1.4 (0.7)	-0.1 (0.8)	0.0 (0.6)	ns	ns
Serial Clustering	0.6 (2.3)	0.4 (2.2)	-0.2 (2.2)	0.9 (2.3)	ns	ns
Primacy recall	0.2 (9.6)	-1.1 (10.3)	-3.3 (12.3)	2.3 (7.5)	ns	ns
Recency recall	2.1 (8.4)	7.1 (14.9)	10.9 (12.2)	1.6 (12.7)	ns	ns
Slope	-0.4 (0.7)	-0.2 (0.5)	-0.6 (0.7)	-0.2 (0.5)	ns	0.04
Consistency	-2.1 (19.2)	-4.5 (9.4)	-8.0 (9.3)	-0.8 (17.0)	ns	ns
Perseveration	0.4 (5.6)	1.4 (7.3)	0.2 (4.6)	1.0 (7.7)	ns	ns
List B versus Trial 1	-10.7 (36.4)	-8.9 (52.1)	-2.9 (39.1)	-13.3 (48.1)	ns	
List A						
List B	-0.5 (1.7)	-0.6 (1.98)	-0.6 (1.7)	-0.6 (1.8)	ns	ns
Wechsler Memory Scale						
Logical memory immediate	0.0 (3.7)	-2.9 (6.2)	-5.9 (5.8)	-0.3 (4.1)	ns	0.003
Logical memory delayed	0.3 (4.2)	-2.1 (5.5)	-3.8 (5.2)	-0.2 (4.2)	ns	0.04
Percent retention	13.2 (39.6)	-0.6 (27.7)	-7.4 (28.0)	10.8 (38.0)	ns	ns
Warrington Recognition Memory Test						
Words	-2.7 (8.1)	-2.9 (6.8)	-4.7 (6.7)	-2.9 (7.4)	ns	ns
Right Anterior Temporal Lobectomy						
Wechsler Memory Scale						
Visual reproduction immediate	0.3 (2.3)	0.5 (1.6)	0.4 (2.5)	0.2 (1.3)	ns	ns
Visual reproduction delayed	-0.5 (3.1)	-0.4 (1.1)	0.0 (3.7)	-9.2 (1.7)	ns	ns
Visual reproduction percent retained	3.3 (33.9)	-19.9 (37.0)	-3.8 (32.3)	-14.2 (31.8)	ns	ns
Warrington Recognition Memory Test						
Faces	-0.05 (5.1)	1.5 (4.6)	-2.4 (4.5)	-1.2 (6.1)	ns	ns
Rey Complex Figure						
Percent retained	-5.7 (18.9)	7.0 (14.3)	-2.0 (25.5)	-0.3 (16.3)	ns	ns
* ns, not significant.						

Table 3. Difference Scores for Extent of Resection and Presence or Absence of Hippocampal Sclerosis^a